

Educational Outcomes & Resource Allocation

Statistical Analysis Report with Charts & Insights

100 Schools · 5 Regions · 3 School Types · May 2026

Executive Summary

This report analyses 100 simulated schools across five geographic regions (North, South, East, West, Central) and three school types (Public, Private, Charter). The objective is to identify which resource and staffing factors most strongly predict student academic performance, measured through average test scores, graduation rates, pass rates, STEM scores, and literacy scores.

85.1 Avg Test Score	82.8% Avg Graduation Rate	\$5,136 Avg Budget / Student	8.4% Avg Dropout Rate
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Summary Statistics

Metric	Mean	Min	Max
Avg Test Score	85.1	35.0	98.0
Graduation Rate (%)	82.8	65.8	99.0
Pass Rate (%)	75.5	30.0	92.4
Dropout Rate (%)	8.4	0.5	14.7
Budget / Student (USD)	\$5,136	\$1,663	\$7,953
Student–Teacher Ratio	24.3	2.2	178.4
Teacher Qualification (%)	74.8	50.6	100.0
Attendance Rate (%)	84.9	70.3	98.5

Correlation Analysis

Pearson correlation coefficients measure linear association between each variable and the primary outcome (Avg Test Score). Bars pointing right (teal) are positive predictors; bars pointing left (coral) are negative predictors. Stronger = longer bar.

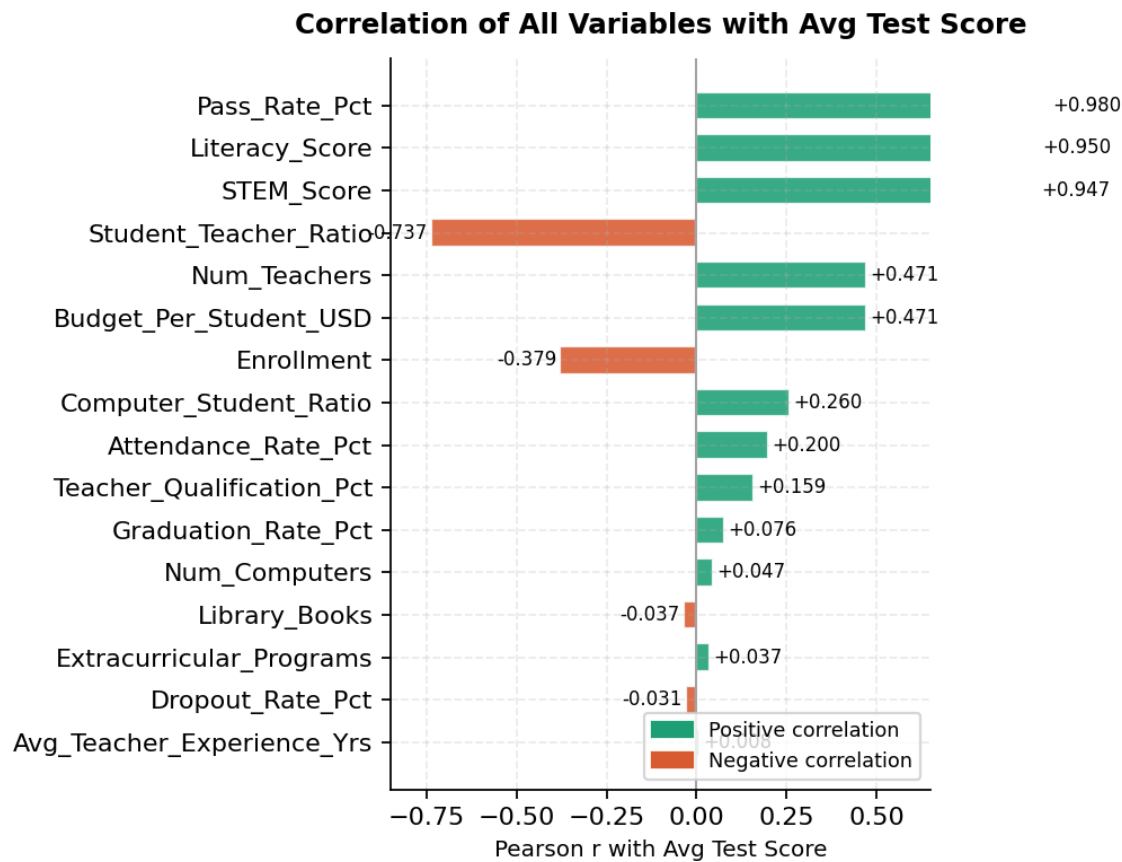


Figure 1. Pearson r of all numeric variables with Avg Test Score, sorted by absolute strength. Student–Teacher Ratio dominates at $r = -0.74$.

Interpretation

- **Student–Teacher Ratio ($r = -0.74$)** — The single strongest predictor. More students per teacher = lower scores.
- **Budget per Student ($r = +0.47$)** — Moderate positive link. Higher spending associates with better performance.
- **Enrollment ($r = -0.38$)** — Larger schools score lower, likely acting as a proxy for higher STR.
- **Attendance Rate ($r = +0.20$)** — Consistent school presence shows a modest positive effect.
- **Teacher Qualification % ($r = +0.16$)** — Surprisingly weak; staffing quantity matters more than credential rate.
- **Extracurriculars & Experience ($r \approx 0$)** — Near-zero; may be mediated through other unmeasured variables.

Pairwise Correlation Heatmap

The heatmap below shows correlations between all key resource and outcome variables. Deep blue = strong positive; deep red = strong negative; white $\approx$ no relationship. Note the near-perfect inverse between Graduation Rate and Dropout Rate (-0.94).

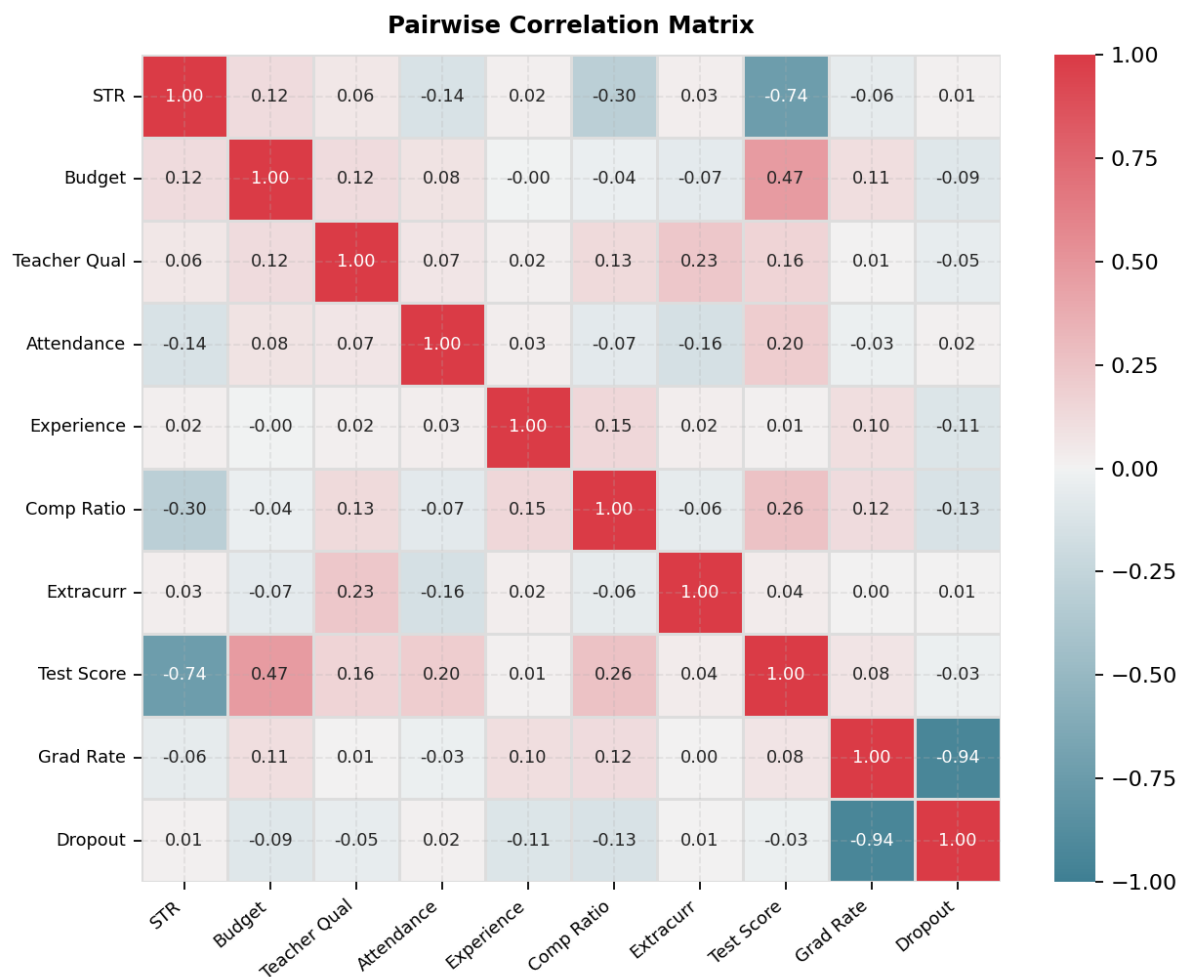


Figure 2. Full pairwise correlation matrix across 10 key variables. The STR–Test Score cell (-0.74) and Grad Rate–Dropout cell (-0.94) are the strongest signals.

Scatter Plots — Resource vs Performance

Each dot represents one school. The dashed OLS line shows the overall trend. These two charts visualise the two strongest resource–outcome relationships in the dataset.

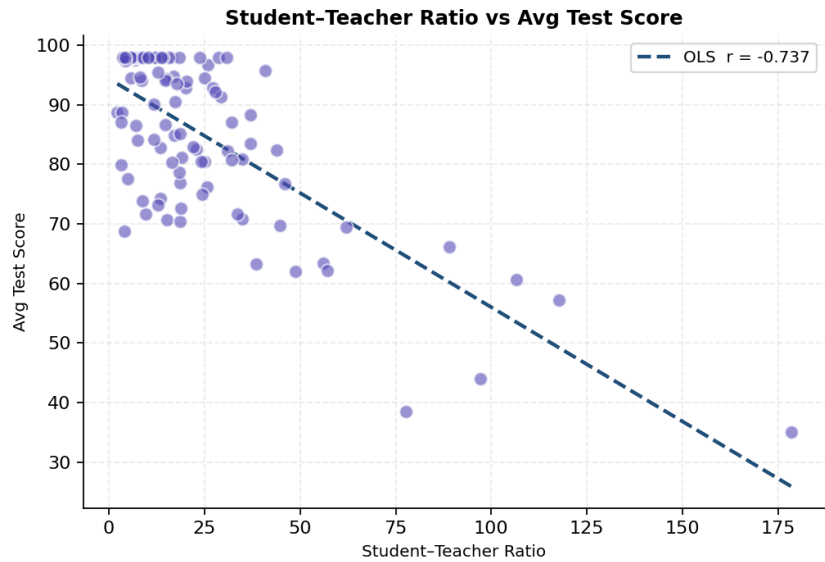


Figure 3. Student–Teacher Ratio vs Avg Test Score ($r = -0.74$, $p < 0.001$). Schools above a ratio of ~40:1 rarely exceed a score of 75.

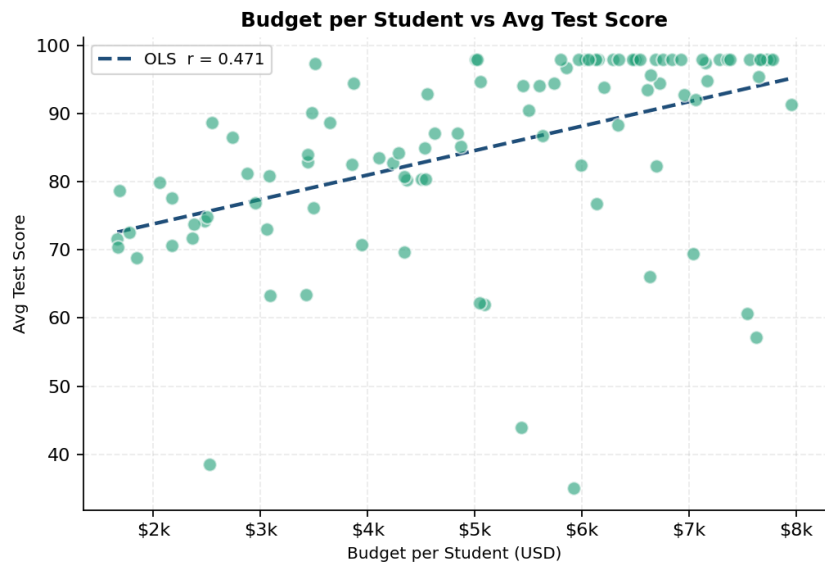


Figure 4. Budget per Student vs Avg Test Score ($r = +0.47$, $p < 0.001$). A positive trend is clear, though variability is wide — budget alone doesn't guarantee outcomes.

Attendance Rate & Budget Tier Analysis

Attendance Rate vs Test Score

Attendance shows a weaker but positive correlation (+0.20). Improving attendance is a lower-cost lever compared to staffing increases.

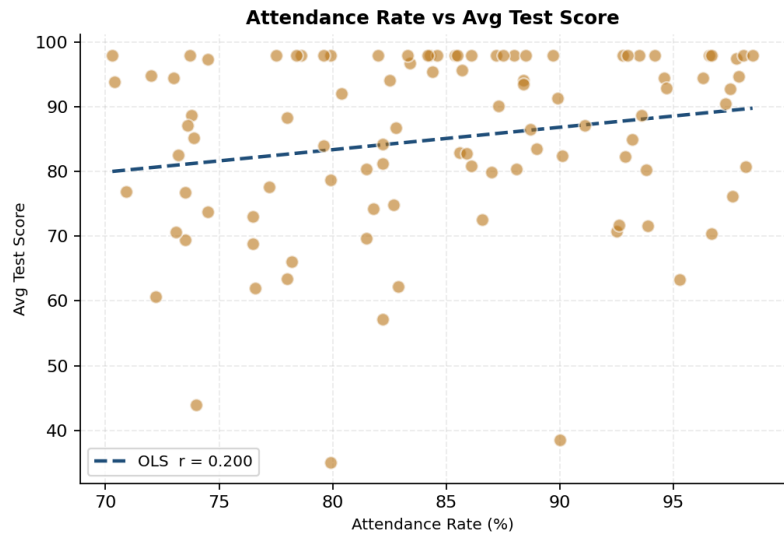


Figure 5. Attendance Rate (%) vs Avg Test Score ($r = +0.20$, $p = 0.048$). A modest but statistically significant positive trend.

Performance by Budget Tier

Schools segmented into four equal-width spending tiers show a clear monotonic pattern — every step up in budget tier delivers higher test scores and pass rates.

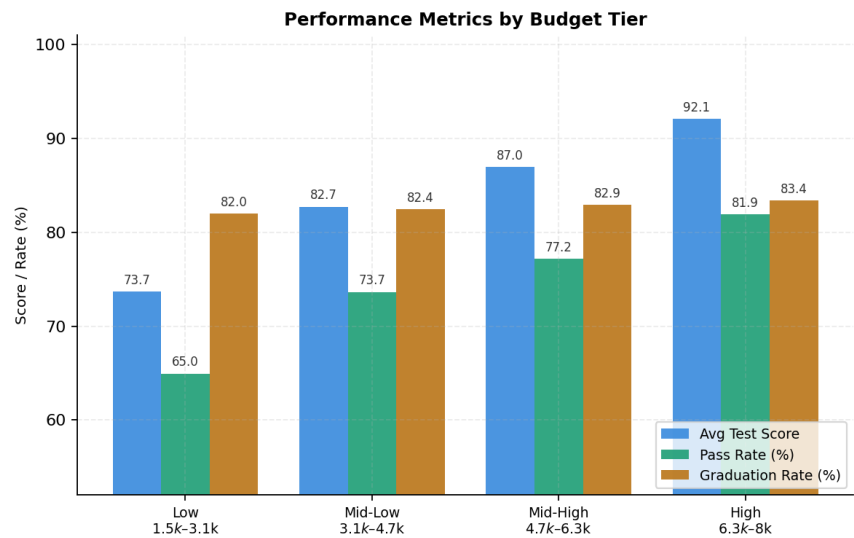


Figure 6. Avg Test Score, Pass Rate, and Graduation Rate by budget tier. High-budget schools (\$6.3k–\$8k) average 92.1 vs 73.7 for low-budget peers — an 18.4-point gap.

Regional & School Type Comparisons

Regional Performance

The South leads in both test scores and budget. The East trails on scores and records the highest dropout rate (9.6%), pointing to a funding equity gap.

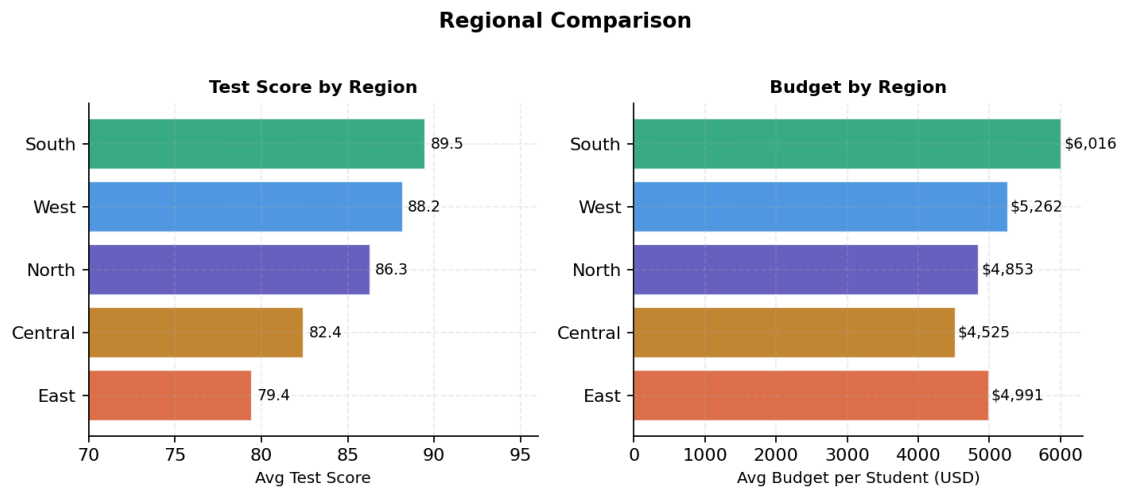


Figure 7. Average test score (left) and budget per student (right) by region. South: score 89.5, budget \$6,016. Central: score 82.4, budget \$4,525.

Performance by School Type

Differences across Public, Private, and Charter schools are modest, confirming that school type is a much weaker predictor than staffing ratio or budget level.

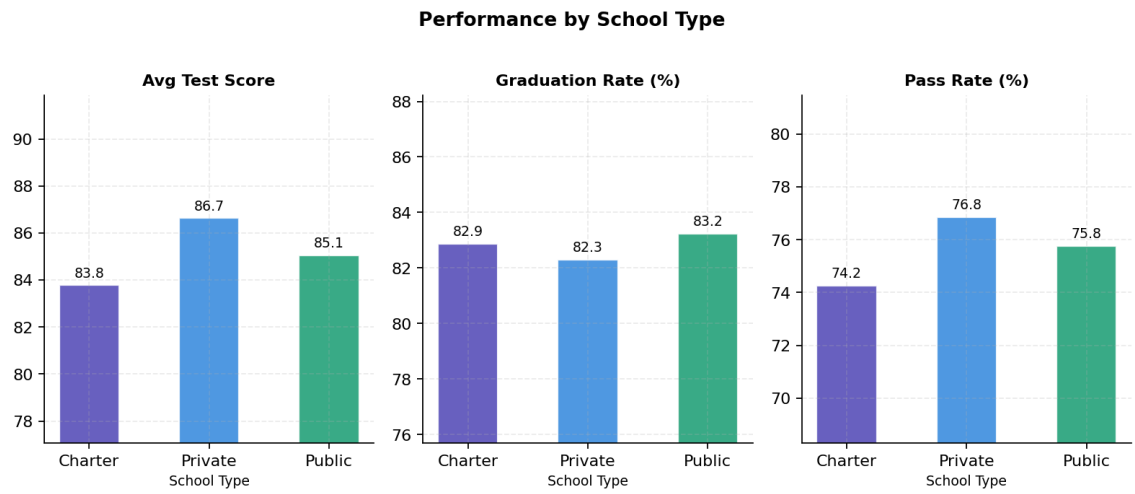


Figure 8. Test score, graduation rate, and pass rate by school type. Private edges on test score (86.7); Public leads on graduation rate (83.2%).

Six Key Insights

1. Student-Teacher Ratio Is the #1 Driver

Correlation of -0.74 — by far the strongest signal in the entire dataset. Schools with ratios above 40:1 consistently fall below 70 points. Every 10-student reduction per teacher associates with $\sim 7\text{--}9$ extra score points. Policy priority: hire more teachers before upgrading infrastructure.

2. Budget Matters — But Allocation Determines ROI

High-budget schools score 18.4 pts above low-budget peers. However, budget correlates only $+0.11$ with graduation rate, suggesting funds channelled into staffing yield a stronger return than capital or technology expenditure.

3. South Leads; East Trails in Both Score & Equity

South records the highest avg test score (89.5) and budget (\$6,016). East has the lowest score (79.4) and the highest dropout rate (9.6%). Regional funding redistribution warrants urgent attention from district administrators.

4. Teacher Credentials Are a Surprisingly Weak Predictor

Teacher qualification % correlates only $+0.16$ with test scores. Schools with 100% credentialled staff do not consistently outperform those at 50%. Staffing headcount outweighs credential level — in-service training may deliver more value than entry gatekeeping.

5. Graduation & Dropout Are Near-Perfect Inverses ($r = -0.94$)

These two metrics capture essentially the same dynamic. Retention strategies and graduation support programmes can be designed as one unified intervention rather than two separate initiatives.

6. Extracurriculars & Experience Show Negligible Direct Links

Extracurricular count ($r = 0.04$) and teacher experience ($r = 0.01$) show near-zero correlations with test scores. Their benefits likely operate through indirect pathways — student engagement, morale, culture — not captured in this snapshot dataset.

Recommendations for Further Investigation

- **Multiple linear regression** — control for region and school type simultaneously to isolate the net effect of each resource variable.
- **Longitudinal analysis** — track whether budget increases at the same school lead to measurable score improvements over time. Cross-sectional data cannot confirm causation.
- **Non-linear STR thresholds** — test whether the negative effect of student–teacher ratio plateaus above 30:1, or keeps worsening linearly, using segmented regression.
- **Socioeconomic controls** — add median household income and free-lunch programme participation rates to disentangle resource effects from socioeconomic confounders.
- **Cluster analysis** — identify school archetypes (e.g., high-budget/low-ratio stars vs. low-budget/high-ratio at-risk schools) and tailor policy interventions by cluster.
- **STEM vs Literacy gap** — school-type and regional variation in the STEM–literacy score gap warrants curriculum-level investigation.

Dataset: 100 simulated schools · Analysis: Python (pandas, scipy, matplotlib, seaborn, reportlab) · All 8 charts embedded as high-resolution figures · May 2026